

Priyanka Chatterjee

Agentic AI & GenAI Engineer · Python · React/TypeScript · MCP · LLM Agents · 4+ Years

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SUMMARY

Full-stack Agentic AI and GenAI engineer with 4+ years building enterprise-grade LLM applications, **multi-agent** workflows, retrieval-augmented generation pipelines, and real-time **streaming** copilot interfaces in **Python** and **React/TypeScript**. Built and shipped an **MCP** server providing standardized, real-time LLM agent access to enterprise cloud data; designed **multi-agent** workflows orchestrating **LLM agents** to generate, validate, and refine policy commands with self-correction loops. Architected **RAG-based Pydantic AI** agent ingesting documentation and code graphs into vector storage, deployed to **AWS Lambda** with Slack Bot API integration for instant developer onboarding. Strong expertise in full-stack AI: **Python** backends (FastAPI-style REST APIs, event-driven architectures), **React/TypeScript** frontends (**WebSocket streaming**, component SDKs), **prompt engineering**, tool/function calling, **responsible AI** guardrails, and production infrastructure (**Docker**, **Kubernetes**, **GitHub Actions**).

TECHNICAL SKILLS

AI & LLM Engineering: LLM Application Development, Agentic Workflows, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Prompt Engineering, Tool/Function Calling, Model Context Protocol (MCP), OpenAI Agents SDK, Pydantic AI, LLM Observability, Semantic Search, Vector Embeddings

Languages: Python, TypeScript, JavaScript (ES6+), Go, SQL, Bash, HCL

Frontend & UI Development: React, TypeScript, Next.js, MUI, WebSocket, Server-Sent Events (SSE), streaming UI, real-time interfaces, component-driven architecture, reusable component SDKs, state management, API-driven UI

Backend & APIs: REST APIs, gRPC, Python backend services, Microservices, Event-Driven Architecture, Serverless, Express.js, Gin, OAuth 2.0

Data & Databases: PostgreSQL, Redis, DuckDB, Vector Databases, S3 Vector Storage, Query Optimization, Embeddings, Knowledge Retrieval

Messaging & Streaming: Apache Kafka, consumer groups, dead-letter queues (DLQs), event-driven pipelines, WebSocket streaming, real-time processing

Cloud & Infrastructure: AWS (Lambda, EC2, S3, CloudWatch, RDS), Azure, GCP, Docker, Kubernetes, OpenShift, multi-cloud

DevOps & CI/CD: GitHub Actions, CI/CD pipelines, automated testing, release automation, containerized deployment, monitoring, observability

Responsible AI & Security: PII protection, data privacy, AI security, responsible AI guardrails, prompt injection mitigation, safe tool execution, AI governance, compliance frameworks (CIS, NIST, HIPAA, PCI DSS)

Core CS: Data structures and algorithms, system architecture, API design, performance optimization, distributed systems

WORK EXPERIENCE

Turbot HQ, Inc. Remote

August 2022 – April 2026

Software Engineer I

- Built the Steampipe Model Context Protocol (**MCP**) server end to end in **Python**, providing **LLM agents** with standardized, real-time access to enterprise cloud and security data for retrieval and analysis; enables agentic workflows to interact with live infrastructure.
- Designed a **multi-agent** workflow on the **OpenAI** Agents SDK orchestrating **LLM agents** that generate cloud preventive-policy commands (e.g., blocking public S3 buckets by region), validate them in a live test environment, self-correct failures, and score accounts against internal security metrics.
- Built a **RAG-based** internal **Pydantic AI** agent that resolves cross-team queries by ingesting documentation and code graphs for all internal repositories into S3 vector storage; deployed to **AWS Lambda** and integrated with Slack Bot API, instantly resolving onboarding and cross-team queries for developers.
- Designed and developed a **REST API** microservice in Go using the Gin framework with OAuth 2.0 authentication, containerized with **Docker**, deployed to **AWS EC2**, and served as a gRPC client for internal microservices; handled production traffic for a SaaS platform.
- Built a **Kafka**-based, fire-and-forget event-driven pipeline for policy-enforcement workflows with asynchronous consumer processing and dead-letter queue (DLQ) for fault tolerance; ensured reliability and observability in high-volume data processing.
- Authored compliance and security modules in HCL + SQL covering 1,000+ resource checks across CIS (**AWS**, Azure, GCP), NIST, HIPAA, PCI DSS frameworks used by enterprise customers in finance and healthcare; implemented **responsible AI** practices and security guardrails for cloud governance.
- Led an LLM-driven Claude Code skill automating code refactoring, unit test generation, deployment, and Terraform-driven integration testing; migrated 35+ Azure cloud-governance modules from polling to push-based eventing via Azure Event Grid, cutting latency 70–90%.
- Engineered core features for Tailpipe (Go-based observability CLI on DuckDB + Parquet) lifting log-processing throughput 30% to 1,000+ records per second; authored 11 **AWS** log-source tables with standardized error handling and cross-platform smoke-test pipeline in **GitHub Actions**.

Kredey (Kredey Innovations Pvt. Ltd.) Remote

2021

Full-Stack Website Developer (Intern)

- Built full-stack **React** + Firebase PWA with full-text and semantic search powered by Elasticsearch, supporting real-time data discovery over a NoSQL backend; implemented real-time notifications and responsive UI/UX.

PROJECTS

Lila Tic-Tac-Toe — Real-Time Multiplayer Game (2026)

Full-stack real-time system: **React** 19 + **TypeScript** frontend on Vercel; Go-based Nakama game server with PostgreSQL on Fly.io. Server-authoritative architecture over **WebSocket** transport with real-time state **streaming**, KV-storage matchmaking, optimistic UI with failure recovery, and reconnect-into-active-match reliability. Demonstrates **streaming** interfaces, real-time data synchronization, and distributed systems reliability patterns critical for copilot and agent UIs.

TODO / Tasks REST API (April 2026)

Full-stack **REST API** in **TypeScript** with Bun, Express.js 5, and SQLite (bun:sqlite). JWT authentication, RBAC, prepared-statement CRUD service layer, database migrations, and schema validation with Zod. Demonstrates component-driven API design, typed clients, reusable service patterns, and production-grade backend practices.

General-Purpose AI Chatbot / Agent with MCP (June 2025 – August 2025)

Agentic chatbot framework built with **Pydantic AI** featuring configurable external **MCP** servers via persistent MongoDB-backed registry, JSON config loader with strict validation, per-server failure isolation, native **tool calling**, mid-conversation multi-model switching (**OpenAI**, Gemini, Anthropic, Mistral), conversation-history persistence, sync + **streaming** output, prompt-injection safeguards, and Logfire observability. Demonstrates **multi-agent** orchestration, tool integration, and production-grade LLM application architecture.

EDUCATION

University of Engineering & Management (UEM) Kolkata, West Bengal, India

August 2019 – May 2023

Bachelor of Technology in Computer Science and Engineering

CGPA: 9.28

Relevant Coursework: Object-Oriented Programming, DBMS, Data Structures and Algorithms, Shell Scripting in Linux, Low-Level Design, High-Level Design